verify that a hash H <= 2128 assuming 2128 is the `function(difficulty)` in the steps? Check whether the leading (256-128= 128) 128 bits of H is zero. *Note that all data and computations are in 256 bits*. In the newer version, Ethereum Istanbul, the plan is to use proof of stake (section 12.3.2).

12.3.2 Proof of stake

In proof of stake (POS), the full node with the most at stake or most coins in its account chooses the next block to be added to the chain, which is why it is called proof of stake. The idea is that the node with the most at stake will not act maliciously and risk its stake by forking the network. A round-robin policy is used to avoid monopoly by the node with the most at stake. The transaction fees pay the minter (yes, *minter*, not *miner*) fee, and there is no miner fee, such as in POW. The POS approach is expected to be environment-friendly and efficient.

12.3.3 Byzantine fault-tolerant consensus

Practical Byzantine Fault Tolerance (PBFT) has been proven to tolerate random or Byzantine node failures (including malicious nodes). In PBFT, nodes elect a leader, and that leader adds the next block to the chain. The nodes exchange messages. The messages, along with the saved state, are used to reach a consensus in the presence of random independent faults or bad nodes. The chosen node adds the next block of validated transactions. PBFT is popular in permissioned blockchains such as Hyperledger Fabric.